Component Software Design/Development

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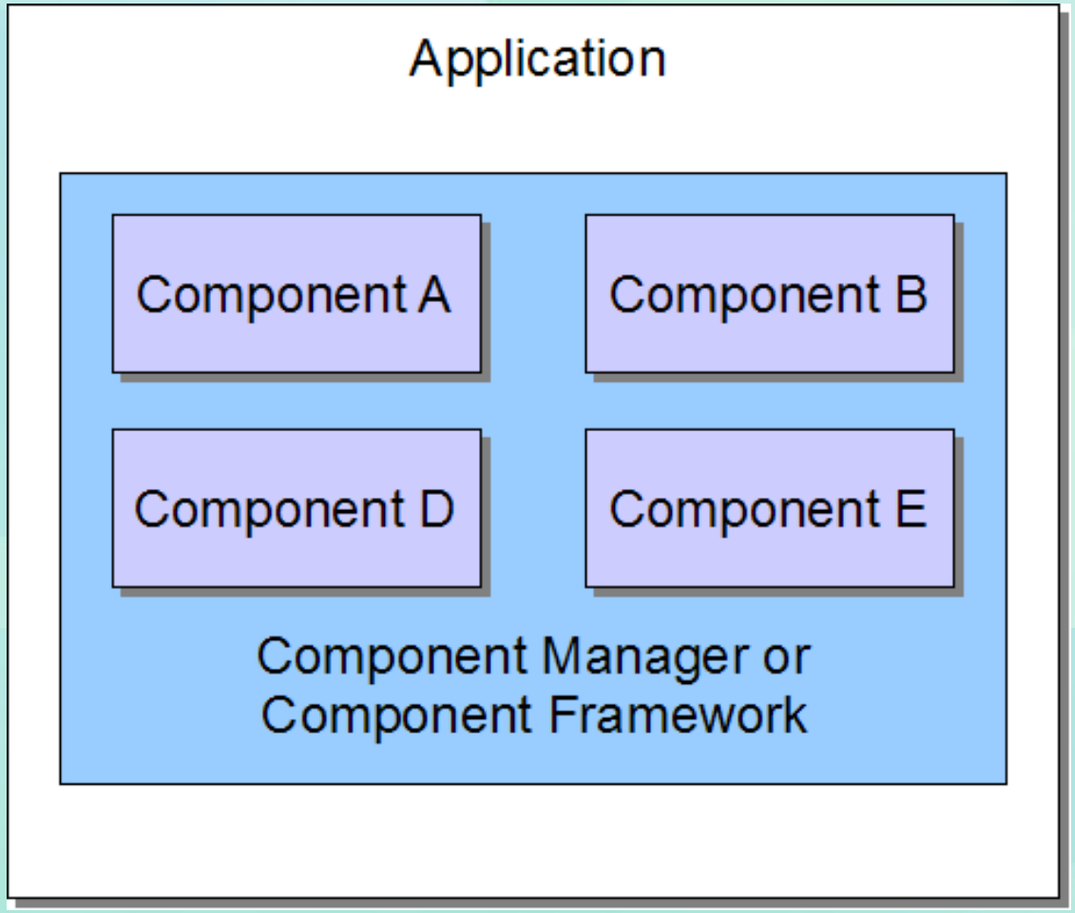
Modern Open Source Software ...

- Is Huge and hence must be modular
- Must provide support for third party functionality
- Must support upgradation in parts
- Must be configurable
- Is appreciated if scriptable

Challenges...

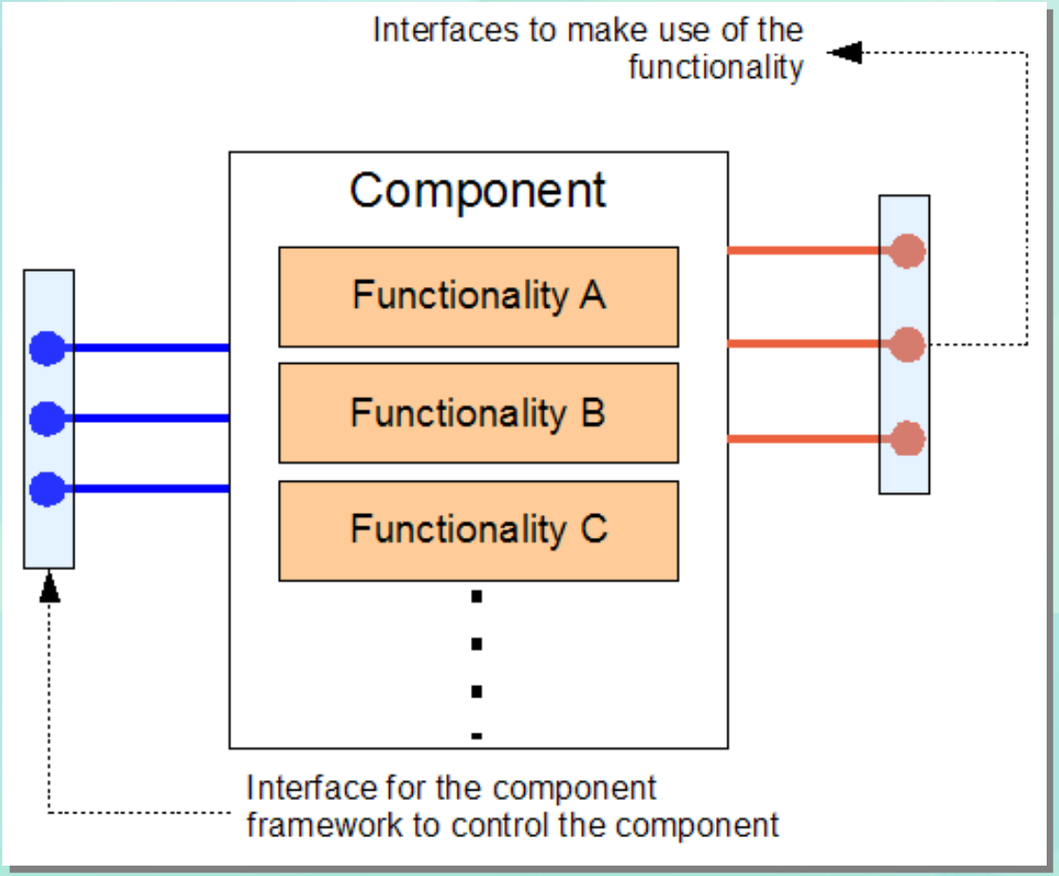
- Separating functionality
- Packaging functionality into loadable components
- GUI Merging
- Inter-Component communication
- Scripting Support

Solution



- Use Component Development Practices.
- Create applications as collection of loosely coupled components.
- Use a component framework to provide GUI Merging and other functionality

Anatomy of a Component



- A component provides
  - Means to query its resource requirements
  - Means to control its state and lifetime.
  - Means for other components to use its functionality

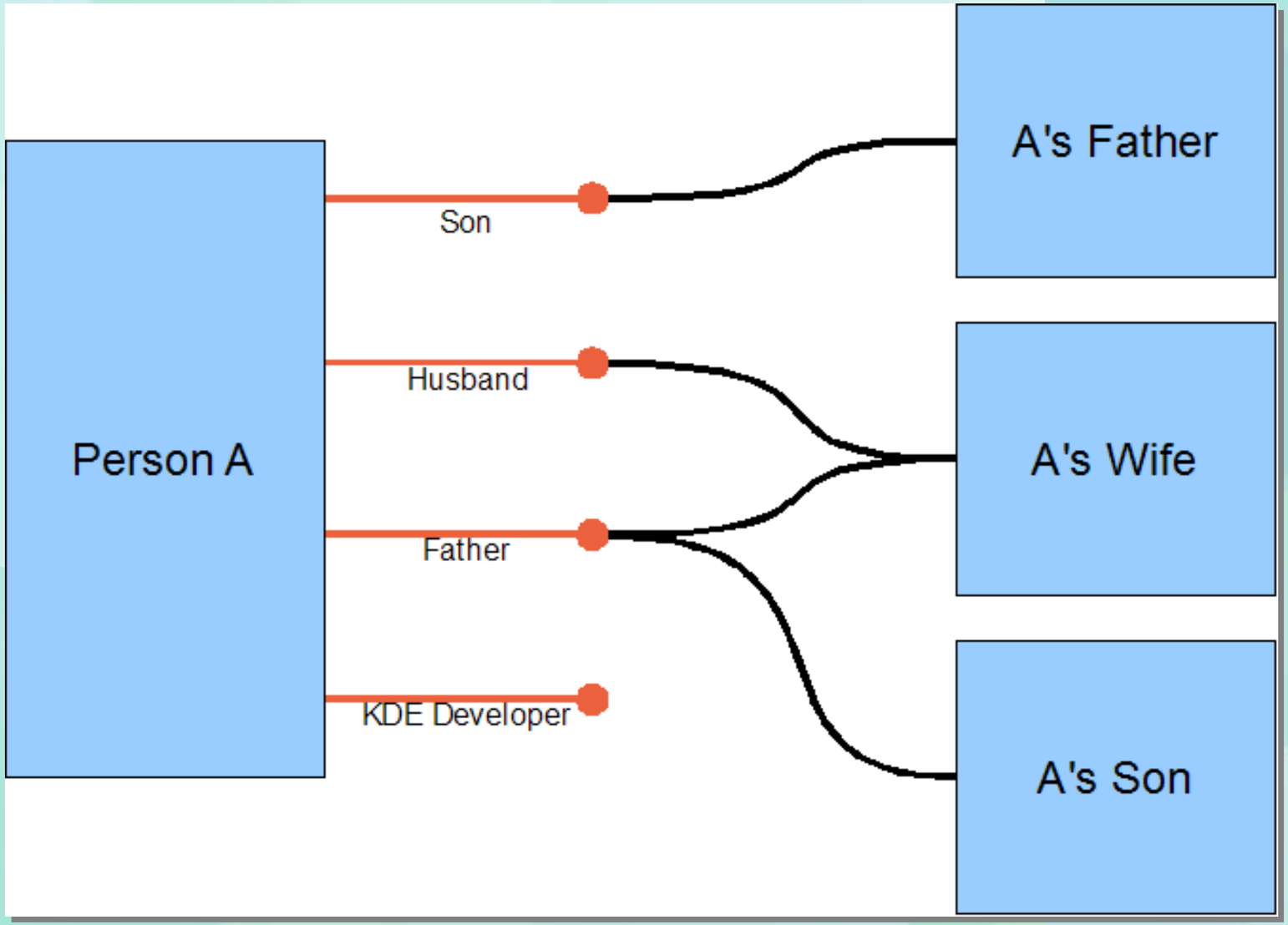
Component Framework

- Provides mechanisms to load components from shared libraries
- Provides mechanisms for components to register and unregister themselves
- Performs GUI Merging on the application container
- Eases inter-component communication

Survey of component frameworks

- CORBA http://www.corba.org
- KParts http://developer.kde.org/documentation/library/3.1-api/kparts/html
- Bonobo http://developer.gnome.org/arch/component/bonobo.html
- COM/DCOM http://www.microsoft.com/com
- GCF http://www.vcreatelogic.com/oss/gcf

Exposing Functionaltity



Qt/C++ Concepts

- Declaring Interfaces
  - Are abstract classes
  - Usually all methods in an interface are declared as pure virtual
  - Marked using the Q_DECLARE_INTERFACE macro
- Implementing Interfaces
  - Virtually inherit one or more abstract classes
  - Mark inherited interfaces using Q_INTERFACES macro
- Querying For Interfaces
  - Done using qobject_cast<> operator

Generic Component Framework

- Foundation for software like
  - VTK Designer 2 – Open Source
  - SequelGUI Circuit Simulator – Closed Source Commercial
- Recognition
  - Winner of QtCentre Programming Contest 2007
- Bits/Bytes...
  - Written using Qt 4.3, and originally for VTK Designer 2
  - Tested on GNU/Linux and Windows

GCF - Concepts

- Components
  - IComponent
    - Base class for all components
  - IComponentPlugin
    - Helps package a component as a plugin
  - GUI XML File
    - Describes exposed objects and resource requirements
- Component Framework
  - CComponentFactory
    - Creates and manages instances of components
  - CComponentGui
    - Performs GUI Merging based on component descriptions provided in a GUI XML file
  - CComponentLoaderComponent
    - Loads components from shared libraries

GCF – Coding Demo

- Marble Component Demo (On-Stage-Coding-Demo).
- Goals
  - To write a component to show MarbleWidget in VTK Designer 2
  - To write a GUI XML file to embed the component in VTK Designer 2
  - To expose certain scriptable methods in the component
  - Connect to IVisNetworkCanvas component in “VTK Designer 2”.